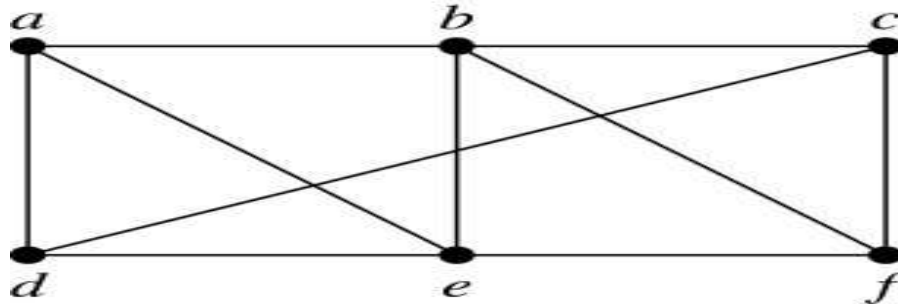


Paths

A **path** is a sequence of edges from one vertex to another vertex. A path will be denoted as a sequences of edges (or vertices in a simple graph).

- The path is a **circuit** if it begins and ends at the same vertex.
- A path or circuit is **simple** if it does not contain the same edge more than once.

Paths (*continued*)



Example: In the simple graph here:

a, d, c, f, e is a simple path of length 4.

d, e, c, a is not a path because e is not connected to c .

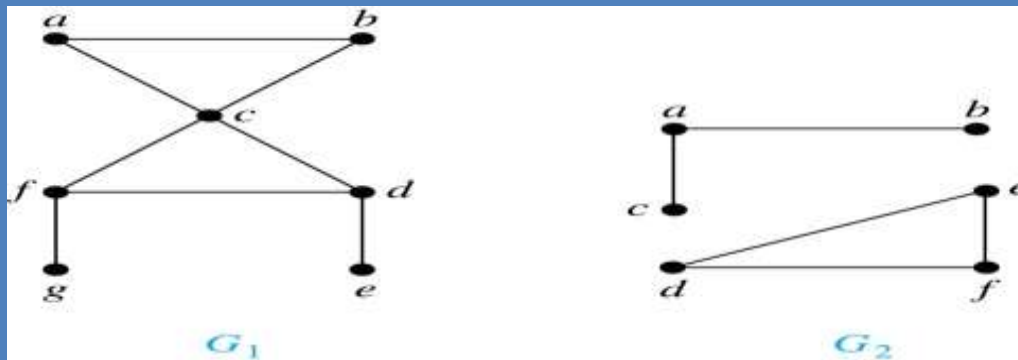
b, c, f, e, b is a circuit of length 4.

a, b, e, d, a, b is a path of length 5, but it is not a simple path.

Connectedness in Undirected Graphs

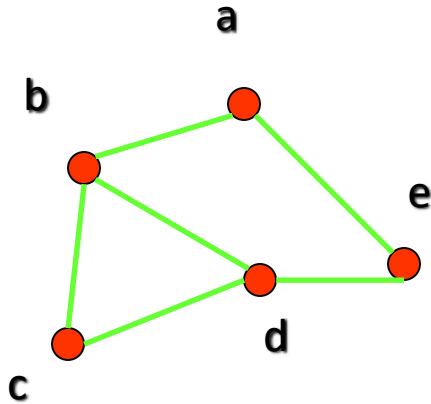
Definition: An undirected graph is called *connected* if there is a path between every pair of vertices. An undirected graph that is not *connected* is called *disconnected*. We say that we *disconnect* a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

Example: G_1 is connected because there is a path between any pair of its vertices, as can be easily seen. However G_2 is not connected because there is no path between vertices a and f , for example.

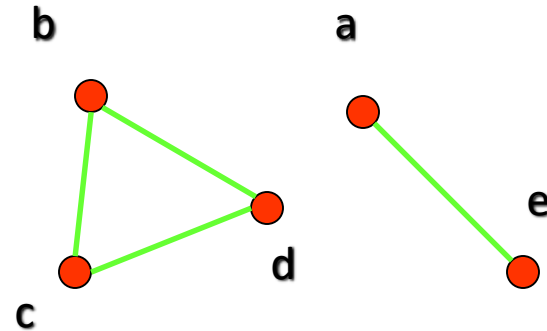


Connectivity

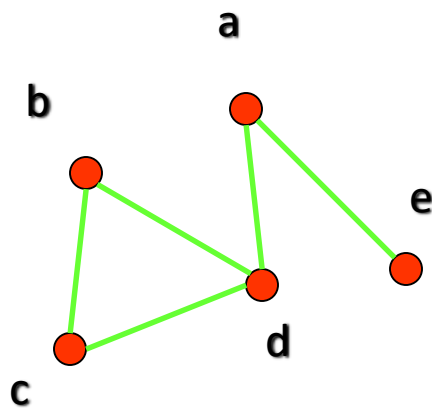
•**Example:** Are the following graphs connected?



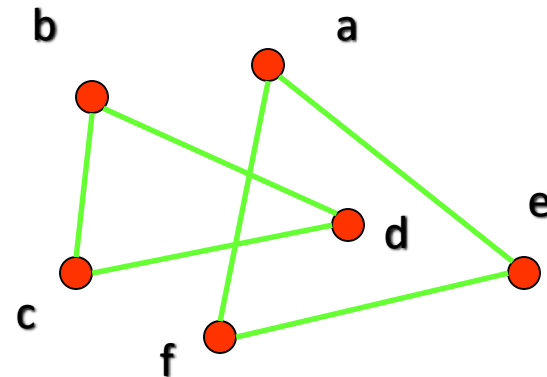
Yes.



No.



Yes.



No.

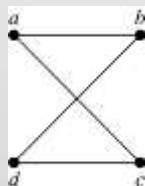
Cut vertices and edges

- If one can remove a vertex (and all incident edges) and produce a graph with more components, the vertex is called a ***cut vertex*** or ***articulation point***.
- Similarly if removal of an edge creates more components the edge is called a ***cut edge*** or ***bridge***.

Counting Paths between Vertices (continued)

Example: How many paths of length four are there from a to d in the graph G .

G



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

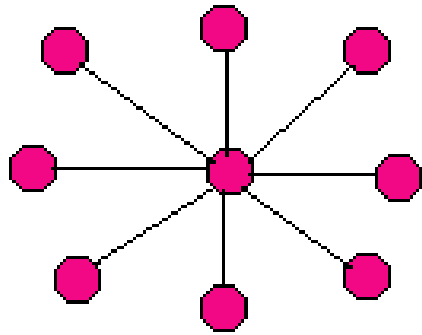
*adjacency
matrix of G*

Solution: The adjacency matrix of G (ordering vertices as a, b, c, d) is given above. Hence the number of paths of length four from a to d is the $(1, 4)$ th entry of $\mathbf{A}^4$. The eight paths are as:

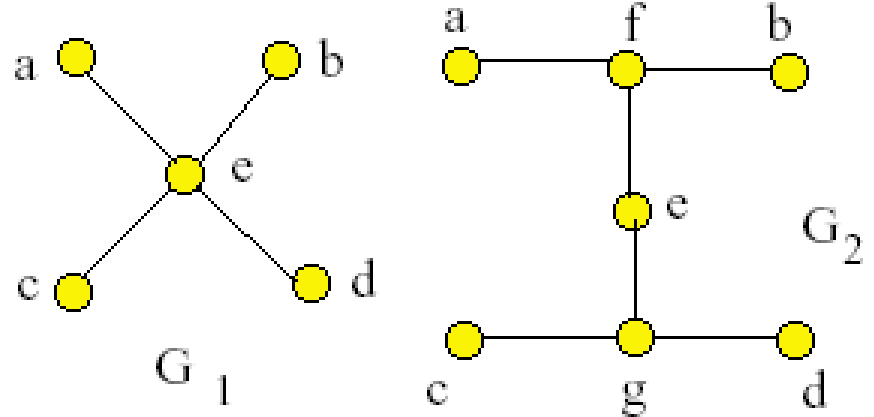
the
the

a, b, a, b, d a, b, a, c, d
 a, b, d, b, d a, b, d, c, d
 a, c, a, b, d a, c, a, c, d
 a, c, d, b, d a, c, d, c, d

$$\mathbf{A}^4 = \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix}$$

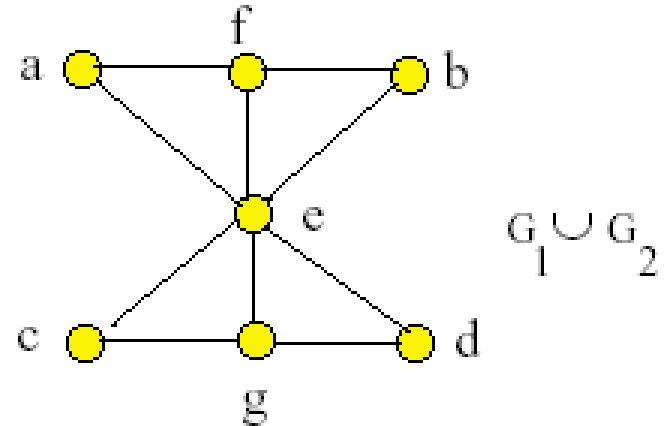


$K_{1,8}$



G_1

G_2



$G_1 \cup G_2$

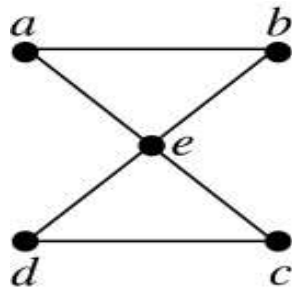
In the star network the
center vertex is a cut vertex.
All edges are cut edges.

In the graphs G_1 and G_2 every edge is a cut edge.
In the union, no edge is a cut edge.
Vertex e is a cut vertex in all graphs.

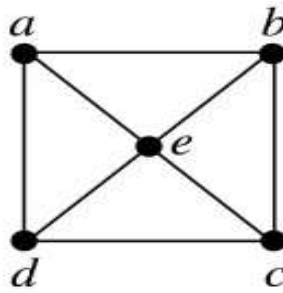
Euler Paths and Circuits

Definition: An *Euler circuit* in a graph G is a simple circuit containing every edge of G . An *Euler path* in G is a simple path containing every edge of G .

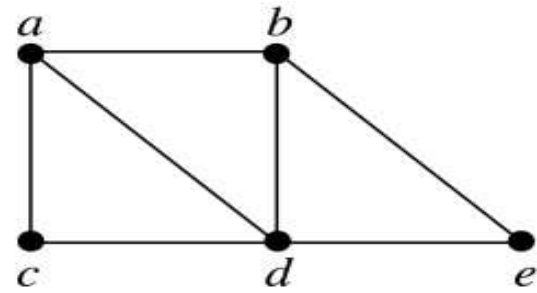
Example: Which of the undirected graphs G_1 , G_2 , and G_3 has a Euler circuit? Of those that do not, which has an Euler path?



G_1



G_2



G_3

Solution: The graph G_1 has an Euler circuit (e.g., a, e, c, d, e, b, a). But, as can easily be verified by inspection, neither G_2 nor G_3 has an Euler circuit. Note that G_3 has an Euler path (e.g., a, c, d, e, b, d, a, b), but there is no Euler path in G_2 , which can be verified by inspection.

Necessary and Sufficient Conditions

- A connected multigraph has an Euler circuit iff *each of its vertices has an even degree*.
- A connected multigraph has an Euler path but not an Euler circuit iff *it has exactly two vertices of odd degree*.

Does this graph have an Euler circuit?

No.

